

1. Project Overview

This report presents the evaluation of multiple regression models developed to predict residential house prices using the House Prices Advanced Regression Techniques dataset. Several machine learning algorithms were trained and compared to identify the most accurate prediction model.

2. Business Objective

The primary objective of this project is to accurately estimate residential property prices based on important housing features. Accurate predictions help buyers, sellers and real estate agencies make informed pricing decisions.

3. Dataset Summary

Attribute	Value
Dataset	House Prices
Total Records	1460
Features	80
Target Variable	SalePrice
Missing Values	Handled
Duplicates	Removed
Encoding	Applied
Scaling	Applied

4. Evaluation Metrics Explanation

MAE measures the average absolute prediction error. Lower values indicate better performance.

RMSE penalizes larger prediction errors more heavily, making it useful for evaluating overall model accuracy.

R² Score indicates how much of the variation in house prices is explained by the model. Values closer to 1 indicate stronger predictive performance.

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Advantages

- High prediction accuracy.
- Handles nonlinear relationships effectively.
- Robust against overfitting compared to a single decision tree.
- Works well with mixed numerical features.

Limitations

- Requires more computation than Linear Regression.
- Larger model size.
- Less interpretable compared to simple regression models.

Conclusion

After evaluating multiple regression algorithms, Random Forest Regressor demonstrated the best balance between prediction accuracy and generalization. Therefore, it was selected as the final deployment model for the Flask application.

Business Recommendations

- Focus on improving property quality before resale.
- Renovating older houses can significantly increase market value.
- Larger living areas and garage spaces positively influence selling prices.
- Use the deployed prediction system as a decision-support tool for pricing.